

Scientific Computation

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Lecture 12 - Errors in Solving Linear Systems

Lecture Outline

- Norms
- Residuals

Vector Norms

A **vector norm** is a function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfies

$$1. \quad \|x\| = 0 \iff x = 0, \quad \Rightarrow \|x\| > 0$$

$$2. \quad \|\alpha x\| = |\alpha| \|x\|, \alpha \in \mathbb{R}, \text{ and}$$

$$3. \quad \|x + y\| \leq \|x\| + \|y\|$$

$\hookrightarrow$ triangular inequality

for all $x, y \in \mathbb{R}^n$.

Think of a norm as an abstract idea of the "size" or "magnitude" of a vector.

Most familiar: Euclidean norm $\|x\|_2 = \left(\sum_{i=1}^n x_i^2 \right)^{\frac{1}{2}}$
 $\hookrightarrow$ "length" of a vector $\mathbb{R}^3$

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p-Norms

The Euclidean norm is also called the 2-norm and can be thought of as a special case of **p-norms** defined as

$$\|\cdot\|_p \triangleq \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}$$



for a real number $p \geq 1$. $\hookrightarrow p=2$ gives us the Euclidean norm

$\rightarrow$ p norm generalizes the Euclidean

$\rightarrow$ 1-norm: $\|x\| \triangleq \sum_{i=1}^n |x_i|$ Manhattan distance

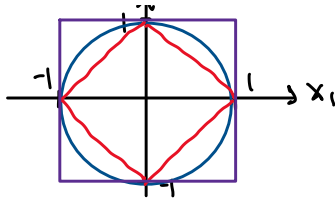
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2-Norm vs. 1-Norm

Let's look at the sets $\|x\|_p \leq 1$ for the Euclidean norm and the 1-norm:

λ_2

$p=2$



$$p=1, \sum_{i=1}^n |x_i| \leq 1$$

$$\hookrightarrow n=2: |x_1| + |x_2| \leq 1$$

$$p=\infty:$$

$$\lim_{p \rightarrow \infty} \left(\sum_{i=1}^n |x_i|^p \right)^{\frac{1}{p}}$$

$$= \lim_{p \rightarrow \infty} \left(\sum_{i=1}^n \left(\frac{|x_i|}{\max(|x_i|)} \right)^p \right)^{\frac{1}{p}} \max(|x_i|)$$

$$= \max_{i=1 \rightarrow n} |x_i|$$

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Matrix Norms

Do we have a similar notion for a matrix $A \in \mathbb{R}^{m \times n}$? Yes, one idea is to have a vector norm $\|\cdot\|$ defined over $\mathbb{R}^m$ and $\mathbb{R}^n$ **induce** the matrix norm

$$\|A\| \triangleq \max_{x \neq 0} \frac{\|Ax\|}{\|x\|} = \max_{\|x\|=1} \|Ax\|.$$

This is also called the **operator norm**, and it has the following familiar properties:

1. $\|A\| = 0 \Leftrightarrow A = 0$
2. $\|\alpha A\| = |\alpha| \cdot \|A\|, \alpha \in \mathbb{R}$
3. $\|A+B\| \leq \|A\| + \|B\|, \forall A, B \in \mathbb{R}^{m \times n} \rightarrow \|A\| - \|B\| \leq \|A-B\|$
4. $\|AB\| \leq \|A\| \cdot \|B\|$
5. $\|A\| \geq 0$

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2-Norm for Matrices

Example 5: What is a useful expression for the matrix 2-norm? We have

$$\|A\|_2 = \max_{\|x\|=1} \|Ax\|_2 \quad \|v\| = \sqrt{v^T v}$$

$$= \max_{\|x\|=1} \sqrt{x^T A^T A x}$$

$$= \max \sqrt{\lambda_i(A^T A)}$$

$i=1 \rightarrow n$ $\hookrightarrow$ i th eigenvalue of $A^T A$

Intuition: we choose x s.t. $\|x\|_2=1$ and it lies along the direction that is **maximally stretched**

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1-Norm and ∞ -Norm

Exercise: Show that the matrix 1-norm is

$$\|A\|_1 = \max_j \sum_{i=1}^n |a_{ij}| \quad (\text{maximum over columns}) \quad \text{column sum}$$

and that the matrix ∞ -norm is

$$\|A\|_\infty = \max_i \sum_{j=1}^n |a_{ij}| \quad (\text{maximum over rows}) \quad \text{row sum}$$

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Comparing Norms

Let's look at how to compute matrix (operator) norms with Julia:

```
using LinearAlgebra
A = [1 3
     2 4]

# Do not use norm(), which treats A like a vector: use opnorm()
println("∞-norm: ${opnorm(A, Inf)}")
println("1-norm: ${opnorm(A, 1)}")
println("2-norm: ${opnorm(A, 2)}")
```

✓ 0.0s

```
∞-norm: 6.0  
1-norm: 7.0  
2-norm: 5.464985704219042
```

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Lecture Outline

FP不知道 x^* (准确值), 用 r 来衡量

- Norms

- Residuals

~~X~~ Note: since we can't know exact solution x^* in computer, so we define residual to determine the difference, and give a upper bound to relative error.

Residuals

Consider the linear system $Ax = b$, and let $\tilde{x}$ be a computed solution (estimate), and let x^* be the exact solution. The relative error is

$$\frac{\|x^* - \tilde{x}\|}{\|x^*\|}, \rightarrow \text{relative error}$$

and we say that the **residual** is:

$$r = b - A\tilde{x} \text{ (or } A\tilde{x} - b)$$

$$r=0 \Leftrightarrow b - A\tilde{x} = 0 \Leftrightarrow A\tilde{x} = b \Leftrightarrow \tilde{x} = x^*$$

In practice, $r \neq 0$ because of FP arithmetic (but we aim to make $\|r\|$ small)

Scaling

Note that $Ax = b$ and $\alpha Ax = \alpha b$ for some $\alpha \neq 0$ have the same solution - but how does this affect the residual for a computed solution $\tilde{x}$?

$$r_\alpha = \alpha b - \alpha A \tilde{x} = \alpha (b - A \tilde{x}) = \alpha r$$

Can be arbitrarily large: $\|r_\alpha\| = \|\alpha r\|$
 $= |\alpha| \cdot \|r\|$

Take away: $\|r\|$ on its own way not be informative

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Residual vs. Error

Example 2: Consider the system $Ax = b$ for

$$A = \begin{bmatrix} 1.2969 & 0.8648 \\ 0.2161 & 0.1441 \end{bmatrix}, \quad b = \begin{bmatrix} 0.8642 \\ 0.1440 \end{bmatrix},$$

and an approximate solution $\tilde{x} = [0.9911, -0.487]^T$.

• The residual is small:

$$r = b - A\tilde{x} \approx [10^{-8}, 10^{-8}]^T, \quad \|r\|_\infty = 10^{-8}$$

• The exact solution is $x^* = \begin{bmatrix} 2 \\ -2 \end{bmatrix}$. The error in $\tilde{x}$ is $x^* - \tilde{x}$

$$= \begin{bmatrix} 1.513 \\ -1.0089 \end{bmatrix}, \quad \|x^* - \tilde{x}\|_\infty = 1.513$$

• Small residual does not imply small error!

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Lecture Outline

- Norms
- Residuals
- **Relative solution error**

Quantifying Relative Solution Error

Given $\tilde{x}$, how large is $\frac{\|x^* - \tilde{x}\|}{\|x^*\|}$?

We can use:

$$r = b - A\tilde{x} = Ax^* - A\tilde{x} = A(x^* - \tilde{x})$$

$$\Rightarrow x^* - \tilde{x} = A^{-1}r$$

$$\|x^* - \tilde{x}\| = \|A^{-1}r\| \leq \|A^{-1}\| \cdot \|r\| \quad (2)$$

Using $b = Ax^*$, $\|b\| = \|Ax^*\| \leq \|A\| \cdot \|x^*\|$

$$\|x^*\| \geq \frac{\|b\|}{\|A\|}$$

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Condition Number

The **condition number** of a matrix A is

$$\text{cond}(A) \triangleq \|A\| \cdot \|A^{-1}\|.$$

From the last slide, we divide the inequalities to get

$$\frac{\|x^* - \tilde{x}\|}{\|x^*\|} \leq \frac{\|A^{-1}\| \|r\|}{\frac{\|b\|}{\|A\|}} = \|A^{-1}\| \|A\| \frac{\|r\|}{\|b\|} = \text{cond}(A) \frac{\|r\|}{\|b\|}$$

$$\therefore \frac{\|x^* - \tilde{x}\|}{\|x^*\|} \leq \text{cond}(A) \frac{\|r\|}{\|b\|}$$

$\Rightarrow$ If $\text{cond}(A)$ is *small*, and $\frac{\|r\|}{\|b\|}$ is *small*,
then our relative error should be small

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Relative Error in Our Example

Example 2 cont.: Returning to the earlier system $Ax = b$, the inverse is

$$A^{-1} = \begin{bmatrix} 1.2969 & 0.8648 \\ 0.2161 & 0.1441 \end{bmatrix}^{-1} = 10^8 \begin{bmatrix} 0.1441 & -0.8648 \\ -0.2161 & 1.2969 \end{bmatrix},$$

and our approximate solution was $\tilde{x} = [0.9911, -0.487]^T$, we had

$$r \approx [10^{-8}, -10^{-8}]^T,$$

and with the 2-norm, $\text{cond}(A) = \approx 2.4973 \times 10^8$.

$$\text{cond}(A) \frac{\|r\|}{\|b\|} \approx 4.0311$$

$$\frac{\|x^* - \tilde{x}\|}{\|x^*\|} \approx 0.6429 \leq 4.0311$$

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在 matrix norm 2 的情况下, condition number 可以化简为 $\|A\|_{2\max}/\|A\|_{2\min}$